

Junwoo Chang

 GitHub |  Homepage |  Google Scholar |  junwoo_chang@berkeley.edu

RESEARCH INTERESTS

Robotics, Reinforcement Learning, Geometric Deep Learning

EDUCATION

Aug. 2026 – Incoming Ph.D. student, **UC Berkeley** (Advisor: Prof. Negar Mehr)
Mar. 2024 – Feb. 2026 M.S., **Yonsei University** (Advisor: Prof. Jongeun Choi)
Mar. 2018 – Feb. 2024 B.S., **Yonsei University**
Leave of absence for mandatory military service (Jul. 2019 – Jan. 2021)

PUBLICATIONS

(*: equal contribution, †: equal advising)

- Jebeom Chae*, **Junwoo Chang***, Seungho Yeom, Yujin Kim, Jongeun Choi
[Multi-Robot Motion Planning from Vision and Language using Heat-Inspired Diffusion](#)
IEEE Robotics and Automation Letters (**RA-L**), 2026
- **Junwoo Chang**, Minwoo Park, Joohwan Seo, Roberto Horowitz, Jongmin Lee†, Jongeun Choi†
[Partially Equivariant Reinforcement Learning in Symmetry-Breaking Environments](#)
International Conference on Learning Representations (ICLR), 2026
- **Junwoo Chang**, Joseph Park, Roberto Horowitz, Jongmin Lee†, Jongeun Choi†
[Group-Invariant Unsupervised Skill Discovery: Symmetry-aware Skill Representations for Generalizable Behavior](#)
Preprint (under review)
- Minwoo Park*, **Junwoo Chang***, Jongeun Choi, Roberto Horowitz
[Symmetry-Aware Steering of Equivariant Diffusion Policies: Benefits and Limits](#)
Preprint (under review)
- Joohwan Seo, Soochul Yoo, **Junwoo Chang**, Hyunseok An, Hyunwoo Ryu, Soomi Lee, Arvind Kruthiventi, Jongeun Choi, Roberto Horowitz
[SE\(3\)-Equivariant Robot Learning and Control: A Tutorial Survey](#)
International Journal of Control, Automation and Systems (IJCAS), 2025
- Hyunwoo Ryu, Jiwoo Kim, Hyunseok An, **Junwoo Chang**, Joohwan Seo, Taehan Kim, Yubin Kim, Chaewon Hwang, Jongeun Choi, Roberto Horowitz
[Diffusion-EDFs: Bi-equivariant Denoising Generative Modeling on SE\(3\) for Visual Robotic Manipulation](#)
Computer Vision and Pattern Recognition (CVPR), 2024 (**Highlight**)
- **Junwoo Chang***, Hyunwoo Ryu*, Jiwoo Kim, Soochul Yoo, Joohwan Seo, Nikhil Potu Surya Prakash, Jongeun Choi, Roberto Horowitz
[Denoising Heat-inspired Diffusion with Insulators for Collision Free Motion Planning](#)
NeurIPS 2023 Workshop on Diffusion Models

AWARDS AND SCHOLARSHIPS

- Best Technical Presentation Award** Oct. 2023
The 5th Yonsei University Mechanical Engineering Graduate Student Academic Conference
- Yonsei Jinri Scholarship** Dec. 2021, Jul. 2022, Dec. 2022
Recognized for sustained academic excellence (three consecutive awards)
- Undergraduate Academic Excellence Honors** 2021
High Honors (Spring 2021), Honors (Fall 2021)

RESEARCH EXPERIENCE

- Machine Learning and Control Systems Laboratory**, Yonsei University Mar. 2024 – Present
Graduate Researcher, Advisor: Prof. Jongeun Choi
 - Research on group equivariant and diffusion-based robot learning
 - Developing partially equivariant reinforcement learning methods for symmetry-breaking tasks
- Machine Learning and Control Systems Laboratory**, Yonsei University Sep. 2022 – Feb. 2024
Undergraduate Research Intern, Advisor: Prof. Jongeun Choi
 - **Hyundai Motor Project:** Self-supervised representation learning for autonomous driving
 - Integrated heat-transfer dynamics with diffusion models for vision-based navigation
 - **Undergraduate Thesis:** *Imaginary Experience Replay: Generating Redundant Transitions for Sparse and Negative Rewards*
- Human-Centered AI Robotics Laboratory**, Yonsei University Jun. 2022 – Sep. 2022
Undergraduate Research Intern, Advisor: Prof. Dongjun Shin
 - Designed a 5-DOF snake robot for in-pipe locomotion and haptic teleoperation

PROJECT EXPERIENCE

- Technical Demonstration of Diffusion-EDFs** Aug. 2023 – Oct. 2023
 - Demonstrated real-world robotic manipulation using the Diffusion-EDFs
 - Awarded Best Technical Demonstration (top project at conference)
- Volunteer Research, Yonsei Rehabilitation Hospital** Jun. 2022 – Jan. 2023
 - Designed assistive systems for children with mental and physical disabilities
 - Developed a collision-aware wheelchair assistant and a posture correction aid for children

TEACHING

- Teaching Assistant**
Mechanical Engineering Laboratory II (Yonsei MEU3005-01), Prof. Jongeun Choi Spring 2024

ACADEMIC SERVICE

- Reviewer, Neural Information Processing Systems (NeurIPS) 2026
- Reviewer, European Conference on Computer Vision (ECCV) 2026
- Reviewer, IEEE International Conference on Robotics and Automation (ICRA) 2026
- Reviewer, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026
- Reviewer, IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM) 2024

MILITARY SERVICE

Republic of Korea Army, K-SAM Pegasus Air Defense System Operator Jul. 2019 – Jan. 2021

SKILLS

Programming	C, C++, Python (PyTorch, JAX, TensorFlow), MATLAB, ROS
Hardware	Franka Emika, Kinova Gen2, RB-Y1, TurtleBot3, OptiTrack, Arduino, Raspberry Pi
Tools	MoveIt, Git, Linux, LaTeX
Theory	Reinforcement Learning, Representation Theory, Group Theory, Diffusion Models